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A Brief Survey on Graph Anomaly Detection

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Abstract

Graph anomaly detection (GAD) has been extensively studied in recent years. GAD aims to detect nodes, edges, and subgraphs that exhibit characteristics and distributions different from those of the majority of graph data. With the advancement of deep learning, many researchers have applied machine learning to address anomaly detection at various scales. In this paper, we classify GAD methods into detector-based and classifier-based approaches and provide a brief introduction and summary of relevant articles from the past three years. Finally, we analyze the challenges and future development directions in the field of GAD.

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Keywords: machine learning; graph Neural Networks; anomaly detection

1. Introduction

Anomalies refer to data samples that show different properties from their counterparts. Those anomalies play important roles in many application domains, such as the detection of financial fraud [27], the monitoring of network instructions [23], and so on. For example, in a financial credit evaluation network, anomalies can be users who forge false identities [14]; in a routing network, anomalies can be fake nodes constructed during hacking; in a logistics network, anomalies can be vehicles or routes that have accidents [9]. Anomalies in the data often bias the analysis, affect researchers' decisions, and cause real-world harm. On the other hand, these abnormal data often have important information for analyzing data and building learning models.

Anomaly detection aims at designing models to distinguish those abnormal samples from the majority of data samples, which has been widely studied in many domains such as computer vision[16], natural language processing[21]. Unlike general regression and classification tasks, anomaly detection faces two main challenges. Firstly, how to de-

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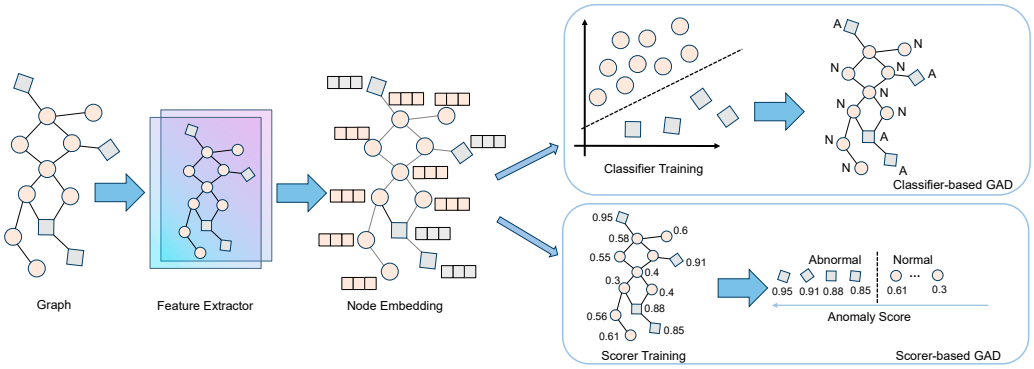


Fig. 1. Illustration of Classifier-based and Scorer-based GAD. After obtaining features of nodes, classifier-based GAD method will train a classifier to determine whether a node is normal (N) or abnormal (A). Differently, Scorer-based GAD will train a scorer and assign anomaly scores to nodes, set thresholds according to score sorting and filter out abnormal nodes.

fine anomalies is challenging. Abnormal data often exhibits complex distributions and characteristics, which makes it difficult to design a general anomaly detection model. Second, a good feature extractor is necessary. Abnormal data in real life is very hidden and often mixed with normal data and difficult to distinguish. A common processing method is to extract and filter features of original attributes to increase the difference between normal data and abnormal data. Therefore, how to design an excellent feature extractor or spatial transformation method is the key to the anomaly detection task. An excellent feature extractor must not only reflect the original attributes of the sample, but also reduce the difficulty of identifying anomalies, which is very difficult in actual operation.

Graph Anomaly detection (GAD) inherits the challenges of general anomaly detection tasks, and will be more difficult due to the complexity of graph data. On the one hand, in addition to its own attributes, graph data also has structural information related to nearby neighbors, which makes it easier for abnormal nodes to be hidden in normal data and makes it more difficult to mine them. On the other hand, there are different scales of anomalies in GAD. Anomaly can be a node, an edge and even a sub graph. In graph datasets, anomaly can be some graphs which are isolated from other graph.

In recent years, thanks to the development of graph machine learning, graph anomaly detection research has also made gratifying progress. In this paper, we summarize the graph anomaly detection models in the past three years and divide them into two categories: **classifier-based models** and **scorer-based models**. The former is based on a supervised learning framework and converts the anomaly detection problem into a binary classification problem. The latter is based on a weakly supervised or unsupervised framework, which calculates anomaly scores and sorts them through methods such as reconstruction and comparison to screen outliers. The illustrations of them are shown in Fig. 1. We summarize and introduce representative methods and provide future development directions.

2. Preliminaries

First, we provide the notations used in this paper. A static graph is denoted as $G(V, E)$, where V is the node set and $E \subset V \times V$ is the edge set. The adjacency and attributed matrices are denoted as $\mathbf{A} \in \mathbb{R}^{n \times n}$ and $\mathbf{X} \in \mathbb{R}^{n \times d}$, respectively, where n is the number of nodes and d is the dimension of the attributes. We also denote the attributed vector for node v_i as $\mathbf{x}_i \in \mathbb{R}^d$. The one-hot label matrix is denoted as $\mathbf{Y} \in \mathbb{R}^{n \times c}$, where c is the number of classes and $y_i \in \{1, 2, \cdots, c\}$ is the label of v_i. A dynamic graph can be formulated as $G(t) = (V(t), E(t))$, where $V(t)$ and $E(t)$ are the node and edge set at timestamp t. Similarly, $X(t)$ is the node attribute matrix at t.

Then we give the following definitions:

Definition 1 (Intra-Graph anomaly detection)**.** *For given graph data,* ***Intra-Graph anomaly detection*** *selects nodes, edges, or subgraphs which show different distribution or characteristic from their counterparts.*

Definition 2 (Graph-level anomaly detection)**.** *Given a graph dataset $\mathcal{G} = \{G_i\}$,* ***Graph-level anomaly detection*** *is to detect anomalous graphs in $\mathcal{G}$.*

In order to give a clear classification of the mentioned methods, we give the following definitions:

Definition 3 (Classifier-based GAD)**.** *Classifier-based graph anomaly detection outputs the predicted label which indicate whether nodes or graphs are anomalous.*

Definition 4 (Scorer-based GAD)**.** *Scorer-based graph anomaly detection return the anomaly score to indicate the probability of samples being anomalous.*

It can be observed that Classifier-based GAD relies on supervised learning framework while Scorer-based GAD can detect anomalies without label information of graphs.

3. Graph Anomaly Detection on Static Graphs

Table 1. Overview of GAD on static graphs.

Name	Model Type	Anomaly Type	Basic Architecture
DONE [1]	scorer-based	nodes	Auto-Encoder
DOMINANT [6]	scorer-based	nodes	Auto-encoder
iGAD [29]	classifier-based	graphs	GCN and Random walk
DAGAD [15]	classifier-based	nodes	GNN
AS-GAD [30]	classifier-based	subgraphs	Auto-Encoder
GHRN [10]	classifier-based	nodes	GNN
OCGTL [22]	scorer-based	graphs	GNN
GLAM [31]	scorer-based	graphs	GIN
SL-GAD [34]	scorer-based	nodes	GNN and Contrastive learning
GraphCAD [5]	classifier-based	nodes	GNN and Contrastive learning
GLocalKD [19]	scorer-based	graph	GCN and Knowledge Distillation
RAND [2]	scorer-based	nodes	Reinforcement Learning, Attention, and Auto-Encoder
GRADATE [7]	scorer-based	nodes	GNN and Contrastive learning
GraphBEAN [8]	scorer-based	nodes and edges	GNN
SparseGAD [11]	classifier-based	nodes	GNN
ACT [26]	scorer-based	nodes	GNN
WEDGE [35]	scorer-based	nodes and graphs	GNN
GOOD-D [17]	scorer-based	graphs	GIN
GmapAD [20]	classifier-based	graphs	GNN and SVM
GAD-NR [24]	scorer-based	nodes	GNN and Auto-Encoders

Classifier-based GAD. Classifier-based methods turn to solving a binary-classification method by introducing supervised label information. Beyond label information, how to obtain features is very important. Many methods use Auto-encoder to reconstruct the graph from the embedding and construct the decision boundary by the reconstruction error. To tackle imbalance problem and anomalous sample scarcity, DAGAD [15] proposes a novel framework based on data augmentation to deal with node anomaly detection. Firstly, DAGAD uses two encoders as information fusion module to obtain node embeddings based on node attributes and graph structures respectively. The samples of the first encoder are concatenated in the original order to form the original samples, and the samples of the second encoder are randomly cross-connected to construct the augmented samples to implement data augmentation module. Finally, DAGAD introduces imbalanced-tailored learning module to use different cross-entropy loss functions to train the original samples and augmented samples separately to improve the accuracy of anomaly detection. For anomaly subgraph anomaly detection, AS-GAE [30] employs an encoder to obtain the node embeddings and a decoder to map the node embeddings to reconstruct the input graph. According to the mismatch between the reconstructed graph and original graph, AS-GAE builds the residual graphs to uncover the anomalous area and uses graph super-modular function to assign anomaly scores to these candidate sub-graphs.

Similarly, to deal with various anomaly notions and imbalance nature in graph-level anomaly detection, iGAD [29] combines attribute-aware graph convolution and random walk kernels to improve the quality of graph embeddings. Specifically, iGAD separately considers the structure information from neighbors at different hops and fuse them to obtain graph convolution embeddings G^{con}. Then iGAD introduces learnable substructures and uses random walk kernels to measure the similarity between the graph and the substructure to obtain graph kernel embeddings G^{ker}. By concatenating them and using a point mutual information-based loss function to train the parameters of GNN and substructures, graph-level anomaly detection can be realized. Some method use supervised contrastive learning to extract features to train a classifier. GraphCAD[5] also introduce contrastive learning to help the detection of anomalous nodes. It introduces a context-aware link predictor to detect suspicious edges and a gumbel-softmax link remover to remove the assumed anomaly nodes to obtain pruned graphs. through a GNN encoder, GraphCAD can obtain the embedding of the global graph, clean graph, and the remaining part. Then contrastive learning is introduced to compute contrastive loss, which constructs positive pairs between global query vector and normal nodes, and negative pairs between query and abnormal nodes. Combining supervised contrastive learning and memory-based mechanism, GraphCAD can train a classifier to detect anomalous nodes.

Abnormal nodes are hidden among normal nodes and are difficult to distinguish, so filtering and sparsifying graph data can help extract abnormalities. GHRN [10] observes that abnormal nodes are sparse so that they are connected to vast normal nodes in anomaly detection task. This paper thinks that heterophilic connections are useless even harmful in GAD, indicating denoising is better than adaptation in this task. It finds that the high-frequency signal part in the Fourier transform of the figure corresponds to high heterophily. So GHRN firstly obtain node embedding from the original graph. Through a low-pass filter, GHRN prunes the graph to delete some heterophily edges. Then GHRN feeds pruned graph into the same encoder to obtain anomalous prediction of nodes and train GNN encoder with cross entropy loss. Similarly, SparseGAD[11] selects useful neighbor information by performing sparsification on graphs and detect anomalous nodes by training a classifier. Specifically, for a given graph, SparseGAD fed it into GNN to obtain the node embeddings and compute the similarity between nodes to obtain 'Homey' matrix. Based on the matrix, SparseGAD performs graph sparsification by filtering unnecessary neighbor nodes and discovering potential neighbor nodes. Then the spasified graph is fed through homey message passing to construct classification loss used to train a anomaly detector.

In addition to train a classifier through deep neural networks, DmapAD [20] paves a new way to perform graph-level anomaly detection. After obtaining the node embeddings produced by GNN models, GmapAD selects the most informative nodes from node sets based on similarity between graphs and nodes and obtain the initial graph mapping. Through differential evolutionary algorithm, new candidate for graph mapping are updated through mutation and crossover. The final graph representations are fed into SVM to detect the anomalous graphs.

Scorer-Based GAD. This kind of method usually extract information from reconstruction and comparison. DONE [1] uses two parallel auto-encoders for link structures and attributes of nodes respectively. Instead of merely using reconstruction loss, DONE introduces three outlier scores for different kind of anomalies to refine the contribution of different nodes to the training of the auto-encoders. Specifically, the larger outlier scores are, the smaller contribution from the nodes will be. By combining the consideration of proximity and homophily property, DONE constructs four different loss to learn the node embeddings and use regularization term to minimize the effect of nodes which have obvious combined outlier scores. The outlier scores trained by auto-encoders can be used for anomaly detection.Similarly, DOMINANT[6] also uses an auto-encoder framework. Firstly, DOMINANT generates node embeddings by an attributed network encoder. Then it employs two decoders to reconstruct the adjacency matrix and attribute matrix respectively and construct loss functions based the difference between the original and reconstructed attributed networks. Finally, the reconstruction error of each node after the model training converges is used as the anomaly score of the corresponding node. GAD-NR [24] detect anomalous graphs by reconstructing different scales of structural information. It introduces a message-passing based GNN model to map nodes into low-dimensional vectors. There are three reconstruction modules in GAD-NR: self reconstruction, node degree reconstruction, and neighbors' representation distribution reconstruction. GAD-NR assumes that the nodes which are hard to be reconstructed are more likely to be anomalous. The reconstruction loss can be directly used as anomalous scores.

Some works demonstrate the effectiveness of contrastive learning. SL-GAD [34] introduces contrastive learning and generative models to detect anomalous nodes in graphs. Firstly, it uses graph view sampling to generate different sub graphs for a certain node. Then two sub graphs are fed into a GNN encoder to obtain node embeddings. On

the one hand, those embeddings are merged into two graph-level representations and contrastive loss is introduced to measure the difference between two augmented views to detect anomalies. On the other hand, GNN decoder is introduced to reconstruct the augmented sub graphs and compute generative loss to measure the structural anomalies. Combining the modules above, SL-GAD construct the overall loss to compute the anomaly scores of nodes. GRADATE [7] uses multi-scale contrastive learning to compute the anomaly scores of nodes. Specifically, GRADATE generates augmented graph view by edge modification and obtains subgraphs using random walk with restart. Then GARDATE design three scales of contrastive learning module: node-node, node-subgraph, and subgraph-subgraph contrast. GRADARE construct contrastive losses mentioned above to train GNN models and the anomaly score is computed according to the similarity between nodes and its negative and positive pairs. GOOD-D [17] consider the out-of-distribution(OOD) anomalies in attributed graphs. It designs a perturbation-free augmentation to protect the distribution of the original graph datasets. GOOD-D uses hierarchical graph contrastive learning to obtain the node embedding considering different scales: node-level contrast and graph-level contrast to obtain graph embedding and group-level contrast to catch the community information. Similarly, anomaly scores are computed according to the score estimator trained by contrastive loss.

Some methods use hypersphere to distinguish anomaly graphs. OCGTL [22] use $K + 1$ GNN encoders to detect anomalous graphs. The loss functions of OCGTL consists of two parts. Graph transformation learning term encourages the node embeddings obtained from other K GNN encoders to be similar to the embeddings of reference GNN encoders. The one class term aims to map data into a minimal hyper-sphere encompassing all normal training data. Based on the loss functions above, OCGTL can train these $K + 1$ GNN encoders and computes the combination those two loss function as anomaly scores.

And some methods improve the distinguish ability through extract **multi-scale** information such as subgraphs, distributions, and so on. GLAM [31] introduce unsupervised GNNs to deal with graph-level anomaly detection. It uses GIN models to generate node embeddings of each graph. For point anomaly graphs where anomalous nodes exist, GLAM uses mean pooling to capture graph embeddings which have been affected by anomalous nodes in message passing framework. For distribution anomalous graph which has different embedding distribution from those of original graphs, Maximum Mean Discrepancy(MMD) pooling is introduced to measure distribution distance between original and anomalous graphs. Interestingly, it obtain the graph embedding by the decomposition of the MMD matrix. An one-class classifier based on hypersphere is designed to select anomalies by computing the distanced-based score. In terms of unsupervised model selection, the paper construct a bi-partite graph to compute the reliability score of a model for certain graphs and select the best models to deal with them without label information. ACT[26] constructs a cross-domain graph anomaly detection model. Firstly, it uses an unsupervised contrastive learning module to learn the initial embedding of normal nodes. Then an anomaly-aware one-class domain alignment is used to align the representation of normal nodes and original graphs and make the abnormal node representation significantly deviate from the normal node representation. Interestingly, ACT introduces self-labelling based deviation learning to label the normal nodes with lowest anomaly scores and iteratively updates parameters of GNN and scoring functions to better the detection process. WEDGE[35] introduces a graph encoder and score estimator to obtain the node scores. From the subgraph level, WEDGE sample subgraphs and compute subgraph scores by assigning attention weight to nodes in corresponding graphs. WEDGE introduces labels of subgraphs to perform subgraph-level deviation loss and introduces contrastive learning to measure the distance between normal and anomalous subgraphs. Based on the loss functions above, WEDGE train the node score estimator to assign anomaly probability to nodes.

In addition to traditional deep learning method, some researchers attempt to apply other learning models into GAD task and deal with different kinds of graph data. GLocalKD [19] introduces knowledge distillation to detect anomalous graphs. It uses a random target network and a predictor network to obtain the node representation and graph representations, respectively. According to the mean square loss of embedding obtained by the two networks from the level of nodes and graphs, GLocalKD compute the anomaly scores through learning the global and local normal patterns to detect anomalous graphs. RAND[2] introduces Reinforcement learning to select neighborhood nodes to produce more anomaly-distinguishing representations. Specifically, it introduces reinforcement neighborhood selection to decide which selection strategy can best reflect the difference between normal and anomaly nodes. Based on selected neighborhoods, RAND introduces attention-based anomaly-aware message aggregator to generate node embeddings. Those embeddings are used to reconstruct topology and attribute information of graphs and anomaly scores are computed according to the reconstruction error, which can also be treated the reward of neighborhood selection strategy.

GraphBEAN [8] considers the detection of nodes and edges in heterogeneous bipartite Node-and-Edge-Attributed graphs. It uses a GNN encoder to obtain node embedding. Based on the obtained embedding, GraphBEAN uses feature decoder and structure decoder to reconstruct attribute and topology information, respectively. The node anomaly score is computed based on the reconstruction error of nodes and its connected edges, and edge anomaly score can be obtained by the reconstruction error and prediction of existence probability of the target edges.

4. Graph Anomaly Detection on Dynamic Graphs

Table 2. Overview of GAD on Dynamic Graphs.

Name	Model Type	Anomaly Type	Basic Architecture
NetWalk [28]	scorer-based	nodes	Auto-Encoder
Deepsphere [25]	scorer-based	nodes	Auto-Encoder
OCAN [33]	scorer-based	nodes	Auto-Encoder
AddGraph [32]	scorer-based	edges	GRU and GCN
StrGNN [4]	classifier-based	edges	GRU and GCN
TADDY [18]	classifier-based	edges	Transformers
DYNANOM [12]	scorer-based	timestamp	PageRank
ANOEDGE and ANOGRAPH [3]	scorer-based	edges and graphs	Higher-order Count-Min Sketch

Classifier-based GAD. StrGNN [4] adopts a sub-graph based method to detect anomalous edges. It firstly samples different sub-graphs of different timestamps from K-hop neighbors of the nodes of the edge. Then a GCN model is introduced to generate the node embeddings from different timestamps.Then sort-pooling is employed to select the node embeddings of the most important nodes as the output of graph feature extraction module. Finally, StrGNN use GRU models to integrate the information of each time stamp to obtain the feature representation of the sub-graph induced by the target edge. Binary cross entropy loss is used to train the edge embeddings and improve the distinguishing ability of anomaly edges.

TADDY [18] designs four modules to detect anomalous edges. Edge-based substructure sampling is to generate sub-graphs based on the target edge from different timestamps. Notice that different from StrGNN, TADDY samples sub-graphs using k most important neighbors based on Pagerank vector instead of k nearest neighbors. Spatial-temporal node encoding is to generate node embeddings by considering diffusion-based, distance-based, and temporal information separately. Dynamic Transformer is to integrate node embeddings to generate target edge embedding based on the subgraph. Discriminative Anomaly Detector is to train a binary anomaly detector using negative sampling and Binary Cross Entropy loss.

Scorer-based GAD. GAD on dynamic graphs is more challenging since it must consider time series data. Netwalk [28] uses auto-encoder framework to generate the node embeddings and introduces K-means clustering to compute the anomaly score according to the distance between the node and the closest clustering center. Different from those traditional methods which merely consider the feature of nodes and its neighbors, Netwalk samples network walk and generates walk embedding using clique embedding to guide the training of node embeddings. To deal with dynamic graphs, Netwalk applies vertex reservoir and incremental learning to describe the changing process of dynamic graphs.

DeepSphere [25] uses hypershpere learning to tell apart of anomaly nodes. Specifically, this method uses a LSTM-Autoencoder to train the node embedding of graphs from different timestamps. Interestingly, hypersphere learning layer is introduced to train a certain hypersphere to distinguish normal and abnormal nodes. The loss function is composed of the reconstruction loss and hypersphere learning loss, which make normal nodes concentrated in the hypersphere and abnormal nodes excluded from the hypersphere.

AddGraph [32] uses GCN to obtain the current embedding of nodes in the current graph. Then it uses a local window to obtain the short information hidden in the few timestamps closest to the current one. A GRU model with attention is introduced to combine the short-term and long-term states to get the final node embeddings. The edge anomaly scores are computed by a sigmoid function whose inputs are the embedding of the corresponding nodes to the edges. In terms of training, AddGraph use a margin-based pairwise loss function to maximize the anomaly scores of the generated edges as negative samples and minimize the anomaly scores of the existed edges.

ANOEDGE and ANOGRAPH [3] perform anomaly detection in streaming graphs and introduce Higher-order Count-Min Sketch to make each hash function map multi-dimensional inputs into a tensor. For edge anomaly detection, AnoEdge-G uses a global approach, maintaining a time-decayed H-CMS to evaluate the anomaly score of each edge by calculating the density of expanding dense sub-matrices. AnoEdge-L focuses on local dense sub-matrix detection around current matrix elements. For graph anomaly detection, AnoGraph maintains an H-CMS to store edge counts, resetting the H-CMS matrix and processing new graph edges to evaluate anomaly scores based on dense sub-matrix density. AnoGraph-K selects the top K elements from the H-CMS, finding the densest sub-matrix around each element to determine the graph's anomaly score. All algorithms use a greedy approach to calculate sub-matrix density, defined as the sum of elements divided by the product of row and column counts.

DYNANOM [12] considers another interesting problem in anomaly detection in dynamic graphs. Instead of detecting certain anomalous nodes or edges, DYNANOM aims to detect the timestamps when the anomaly structures or attributes of target nodes start to appear, i.e., anomaly tracking. It incrementally computes the personalize pagerank vectors (PPV) to obtain the embeddings of the target nodes observed. Based on these embeddings. DYNANOM computes the node anomaly scores from different timestamps, which indicates the probability that anomaly changes occur at this timestamp.

5. Challenges and Feature directions

Although research on anomaly detection has made amazing progress, it still faces two challenges, which determine future research directions. Firstly, the datasets and metrics should be enriched. Due to the rarity of anomalous samples, increasing the varieties of datasets for GAD is also a search hotpot recently. DGraph [13] is a dynamic attributed financial graph for fraudsters detection. In this graph, nodes denote users of Finvolution Industry and edges denote the emergency contact relationship. In addition, borrowing behavior and information are introduce as node attributes. Compared with other GAD dataset, Dgraph has a lager size and smaller anomaly rate, which is consistent with reality world graph with anomalies. Second, how to define anomalies still needs to be studied. How to quantify the degree of anomalies and how to design a general feature extractor to distinguish different anomalies are still problems that need to be solved.

6. Conclusion

In this paper, we sort out and summarize the papers related to graph anomaly detection in recent years. We classify all models into classifier-based and scorer-based models, and introduce representative works and techniques among them. Finally, we discuss future research directions and point out that the construction of datasets and the quantification of anomalies are still important research directions.

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References

[1] Bandyopadhyay, S., N, L., Vivek, S.V., Murty, M.N., 2020. Outlier resistant unsupervised deep architectures for attributed network embedding, in: Proceedings of the 13th international conference on web search and data mining, pp. 25–33.

[2] Bei, Y., Zhou, S., Tan, Q., Xu, H., Chen, H., Li, Z., Bu, J., 2023. Reinforcement neighborhood selection for unsupervised graph anomaly detection, in: 2023 IEEE International Conference on Data Mining (ICDM), IEEE. pp. 11–20.

[3] Bhatia, S., Wadhwa, M., Kawaguchi, K., Shah, N., Yu, P.S., Hooi, B., 2023. Sketch-based anomaly detection in streaming graphs, in: Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 93–104.

[4] Cai, L., Chen, Z., Luo, C., Gui, J., Ni, J., Li, D., Chen, H., 2021. Structural temporal graph neural networks for anomaly detection in dynamic graphs, in: Proceedings of the 30th ACM international conference on Information & Knowledge Management, pp. 3747–3756.

[5] Chen, B., Zhang, J., Zhang, X., Dong, Y., Song, J., Zhang, P., Xu, K., Kharlamov, E., Tang, J., 2023. Gccad: Graph contrastive coding for anomaly detection. IEEE Transactions on Knowledge and Data Engineering 35, 8037–8051.
[6] Ding, K., Li, J., Bhanushali, R., Liu, H., 2019. Deep anomaly detection on attributed networks, in: Proceedings of the 2019 SIAM International Conference on Data Mining, SIAM. pp. 594–602.
[7] Duan, J., Wang, S., Zhang, P., Zhu, E., Hu, J., Jin, H., Liu, Y., Dong, Z., 2023. Graph anomaly detection via multi-scale contrastive learning networks with augmented view, in: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 7459–7467.
[8] Fathony, R., Ng, J., Chen, J., 2023. Interaction-focused anomaly detection on bipartite node-and-edge-attributed graphs, in: 2023 International Joint Conference on Neural Networks (IJCNN), IEEE. pp. 1–10.
[9] Fernandes, G., Rodrigues, J.J., Carvalho, L.F., Al-Muhtadi, J.F., Proença, M.L., 2019. A comprehensive survey on network anomaly detection. Telecommunication Systems 70, 447–489.
[10] Gao, Y., Wang, X., He, X., Liu, Z., Feng, H., Zhang, Y., 2023. Addressing heterophily in graph anomaly detection: A perspective of graph spectrum, in: Proceedings of the ACM Web Conference 2023, pp. 1528–1538.
[11] Gong, Z., Wang, G., Sun, Y., Liu, Q., Ning, Y., Xiong, H., Peng, J., 2023. Beyond homophily: Robust graph anomaly detection via neural sparsification, in: Proceedings of International Joint Conference on Artificial Intelligence, pp. 2104–2113.
[12] Guo, X., Zhou, B., Skiena, S., 2022. Subset node anomaly tracking over large dynamic graphs, in: Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 475–485.
[13] Huang, X., Yang, Y., Wang, Y., Wang, C., Zhang, Z., Xu, J., Chen, L., Vazirgiannis, M., 2022. Dgraph: A large-scale financial dataset for graph anomaly detection. Advances in Neural Information Processing Systems 35, 22765–22777.
[14] Huang, Z., Zheng, H., Li, C., Che, C., 2024. Application of machine learning-based k-means clustering for financial fraud detection. Academic Journal of Science and Technology 10, 33–39.
[15] Liu, F., Ma, X., Wu, J., Yang, J., Xue, S., Beheshti, A., Zhou, C., Peng, H., Sheng, Q.Z., Aggarwal, C.C., 2022. Dagad: Data augmentation for graph anomaly detection, in: 2022 IEEE International Conference on Data Mining (ICDM), IEEE. pp. 259–268.
[16] Liu, J., Xie, G., Wang, J., Li, S., Wang, C., Zheng, F., Jin, Y., 2024. Deep industrial image anomaly detection: A survey. Machine Intelligence Research 21, 104–135.
[17] Liu, Y., Ding, K., Liu, H., Pan, S., 2023. Good-d: On unsupervised graph out-of-distribution detection, in: Proceedings of the Sixteenth ACM International Conference on Web Search and Data Mining, pp. 339–347.
[18] Liu, Y., Pan, S., Wang, Y.G., Xiong, F., Wang, L., Chen, Q., Lee, V.C., 2021. Anomaly detection in dynamic graphs via transformer. IEEE Transactions on Knowledge and Data Engineering .
[19] Ma, R., Pang, G., Chen, L., van den Hengel, A., 2022. Deep graph-level anomaly detection by glocal knowledge distillation, in: Proceedings of the fifteenth ACM International Conference on Web Search and Data Mining, pp. 704–714.
[20] Ma, X., Wu, J., Yang, J., Sheng, Q.Z., 2023. Towards graph-level anomaly detection via deep evolutionary mapping, in: Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 1631–1642.
[21] Park, C.H., 2023. A comparative study for outlier detection methods in high dimensional text data. Journal of Artificial Intelligence and Soft Computing Research 13, 5–17.
[22] Qiu, C., Kloft, M., Mandt, S., Rudolph, M., 2022. Raising the bar in graph-level anomaly detection, in: Raedt, L.D. (Ed.), Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence, IJCAI 2022, Vienna, Austria, 23-29 July 2022, ijcai.org. pp. 2196–2203. URL: https://doi.org/10.24963/ijcai.2022/305, doi:10.24963/ijcai.2022/305.
[23] Ragel, R.G., Parameswaran, S., Kia, S.M., 2005. Micro embedded monitoring for security in application specific instruction-set processors, in: Proceedings of the 2005 international conference on Compilers, architectures and synthesis for embedded systems, pp. 304–314.
[24] Roy, A., Shu, J., Li, J., Yang, C., Elshocht, O., Smeets, J., Li, P., 2024. Gad-nr: Graph anomaly detection via neighborhood reconstruction, in: Proceedings of the 17th ACM International Conference on Web Search and Data Mining, pp. 576–585.
[25] Teng, X., Yan, M., Ertugrul, A.M., Lin, Y.R., 2018. Deep into hypersphere: Robust and unsupervised anomaly discovery in dynamic networks, in: Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence.
[26] Wang, Q., Pang, G., Salehi, M., Buntine, W., Leckie, C., 2023. Cross-domain graph anomaly detection via anomaly-aware contrastive alignment, in: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 4676–4684.
[27] West, J., Bhattacharya, M., 2016. Intelligent financial fraud detection: a comprehensive review. Computers & security 57, 47–66.
[28] Yu, W., Cheng, W., Aggarwal, C.C., Zhang, K., Chen, H., Wang, W., 2018. Netwalk: A flexible deep embedding approach for anomaly detection in dynamic networks, in: Proceedings of the 24th ACM SIGKDD International Conference on knowledge Discovery and Data Mining, pp. 2672–2681.
[29] Zhang, G., Yang, Z., Wu, J., Yang, J., Xue, S., Peng, H., Su, J., Zhou, C., Sheng, Q.Z., Akoglu, L., et al., 2022. Dual-discriminative graph neural network for imbalanced graph-level anomaly detection. Advances in Neural Information Processing Systems 35, 24144–24157.
[30] Zhang, Z., Zhao, L., 2022. Unsupervised deep subgraph anomaly detection, in: 2022 IEEE International Conference on Data Mining (ICDM), IEEE. pp. 753–762.
[31] Zhao, L., Sawlani, S., Srinivasan, A., Akoglu, L., 2022. Graph anomaly detection with unsupervised gnns. arXiv preprint arXiv:2210.09535 .
[32] Zheng, L., Li, Z., Li, J., Li, Z., Gao, J., 2019a. Addgraph: Anomaly detection in dynamic graph using attention-based temporal gcn., in: IJCAI, p. 7.
[33] Zheng, P., Yuan, S., Wu, X., Li, J., Lu, A., 2019b. One-class adversarial nets for fraud detection, in: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 1286–1293.
[34] Zheng, Y., Jin, M., Liu, Y., Chi, L., Phan, K.T., Chen, Y.P.P., 2021. Generative and contrastive self-supervised learning for graph anomaly detection. IEEE Transactions on Knowledge and Data Engineering 35, 12220–12233.
[35] Zhou, Q., Ding, K., Liu, H., Tong, H., 2023. Learning node abnormality with weak supervision, in: Proceedings of the 32nd ACM International Conference on Information and Knowledge Management, pp. 3584–3594.